TCGA Lung Squamous Cell Carcinoma (LUSC) RNA-seq Biomarker Discovery

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Abstract

Cancer transcriptomics enables systematic identification of molecular programs associated with tumor development and patient outcome. In this study, we performed a reproducible analysis of RNA-seq data from The Cancer Genome Atlas (TCGA) lung squamous cell carcinoma (LUSC) cohort to identify tumor-associated transcriptional signatures and evaluate their prognostic relevance.

Using differential expression analysis (DESeq2), pathway enrichment analysis (Gene Ontology and KEGG), and survival modeling, we identified robust transcriptional differences between tumor and normal lung tissue. Functional enrichment analysis revealed strong activation of epithelial differentiation programs and immune-related pathways characteristic of LUSC biology.

We further derived a multi-gene transcriptomic signature that significantly stratifies patients by overall survival and remains predictive after adjustment for clinical covariates. External validation in an independent dataset supports the reproducibility of the identified signal.

Together, these results demonstrate that transcriptome-derived gene expression programs capture both tumor biology and clinically relevant heterogeneity in lung squamous cell carcinoma.

Introduction

Lung cancer remains one of the leading causes of cancer-related mortality worldwide. Lung squamous cell carcinoma (LUSC) represents a major histological subtype of non-small cell lung cancer characterized by extensive transcriptional dysregulation and complex interactions with the tumor microenvironment.

Large-scale cancer genomics initiatives such as The Cancer Genome Atlas (TCGA) have generated comprehensive molecular datasets that enable systematic characterization of tumor biology. RNA-sequencing data in particular provide an opportunity to identify transcriptional programs associated with tumor development and clinical outcome.

In this study, we performed a reproducible transcriptomic analysis of TCGA-LUSC RNA-seq data to:

- identify differentially expressed genes between tumor and normal lung tissue
- characterize biological pathways associated with LUSC
- construct a transcriptome-derived prognostic gene signature
- evaluate its clinical relevance using survival analysis

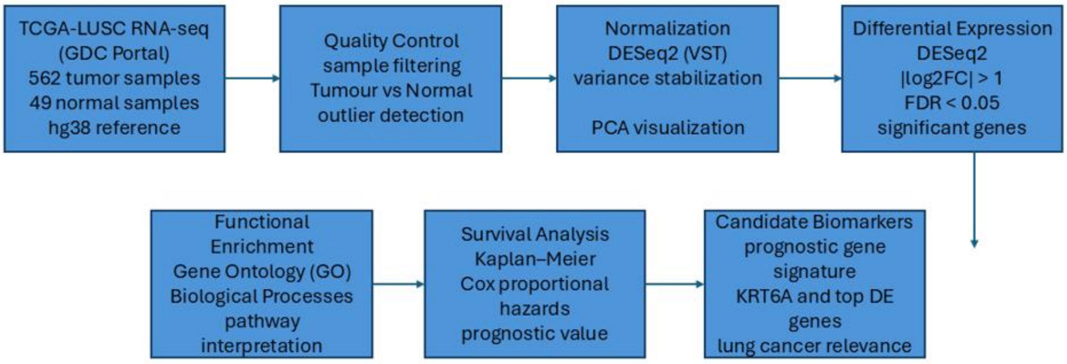

Figure 1. Analysis workflow of the TCGA-LUSC RNA-seq pipeline.

Overview of the reproducible workflow including data acquisition, differential expression analysis, pathway enrichment, and survival modeling.

Methods

Data acquisition

RNA-sequencing gene expression data for lung squamous cell carcinoma (LUSC) were obtained from The Cancer Genome Atlas (TCGA) via the Genomic Data Commons (GDC) portal. Data were downloaded programmatically using the TCGAbiolinks R package.

The analyzed cohort consisted of primary tumor samples (TP) and matched normal lung tissue samples (NT). In total, the dataset included approximately 511 tumor samples and 51 normal samples, with gene expression counts available for ~60,000 genes.

Raw count matrices and associated clinical metadata were retrieved using the TCGAbiolinks interface and processed within the R statistical environment.

Data preprocessing and quality control

Raw RNA-seq count data were imported into R and subjected to standard quality control procedures. Library size distributions and sample correlations were examined to detect potential outliers.

Lowly expressed genes were filtered prior to downstream analyses. Gene expression normalization was performed using the DESeq2 variance stabilizing transformation to account for differences in sequencing depth and RNA composition.

Principal Component Analysis (PCA) was performed on the most variable genes to assess global transcriptomic structure and evaluate separation between tumor and normal samples.

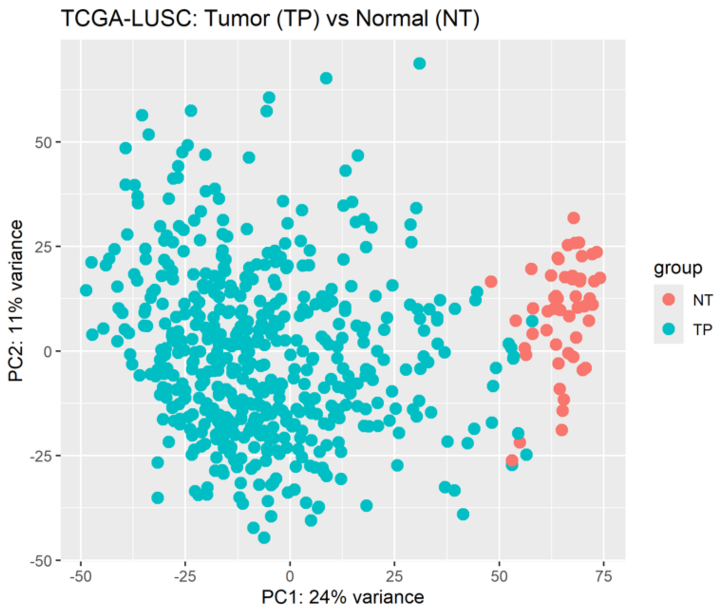

Figure 2. Principal component analysis of TCGA-LUSC samples.

PCA demonstrates clear separation between tumor (TP) and normal lung tissue (NT), indicating large-scale transcriptomic differences between conditions.

Quality control metrics included:

- library size distribution
- sample-to-sample correlation analysis
- PCA visualization of transcriptomic variance

Differential gene expression analysis

Differential gene expression analysis between tumor and normal samples was performed using the DESeq2 package.

The statistical model used a negative binomial generalized linear model to estimate gene-wise dispersion and test for differential expression between experimental conditions. Log2 fold changes were estimated using DESeq2's shrinkage approach.

Genes were considered significantly differentially expressed using the following criteria:

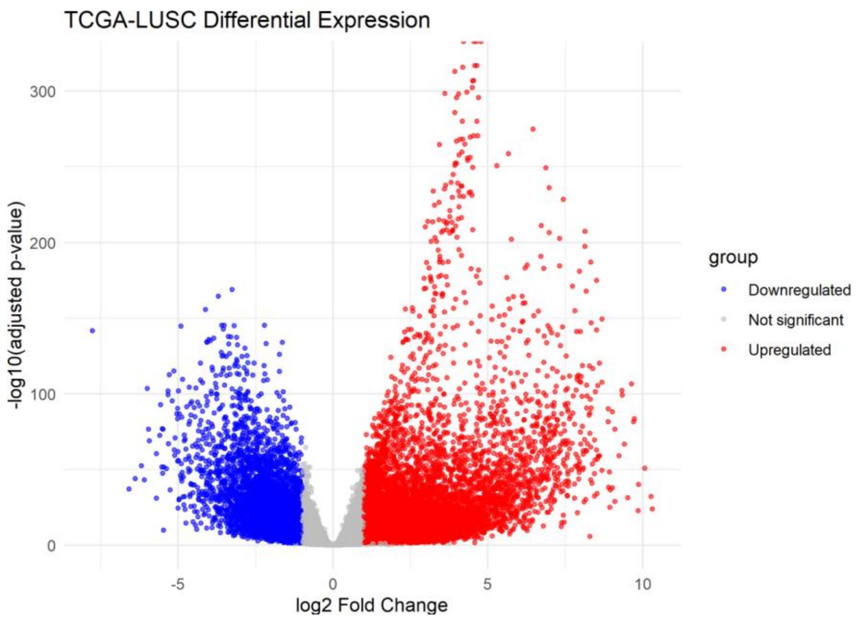

Figure 3. Differential gene expression between tumor and normal lung tissue.

Volcano plot showing significantly upregulated and downregulated genes identified using DESeq2 (FDR < 0.05).

- False Discovery Rate (FDR) < 0.05
- ranking by log2 fold change

Multiple testing correction was performed using the Benjamini–Hochberg procedure.

Significantly dysregulated genes were used for downstream pathway enrichment and biomarker discovery analyses.

Functional enrichment analysis

To interpret the biological significance of differentially expressed genes, functional enrichment analyses were performed using the clusterProfiler R package. Two complementary pathway resources were used:

Gene Ontology (GO)

Gene Ontology enrichment analysis was used to identify overrepresented biological processes among differentially expressed genes. Enrichment was evaluated using hypergeometric testing with multiple testing correction.

KEGG pathway analysis

Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis was performed to identify dysregulated molecular pathways associated with LUSC tumor biology.

Significantly enriched pathways were defined using:

- adjusted p-value (FDR) < 0.05

Enrichment results were visualized using dot plots representing gene ratio and statistical significance.

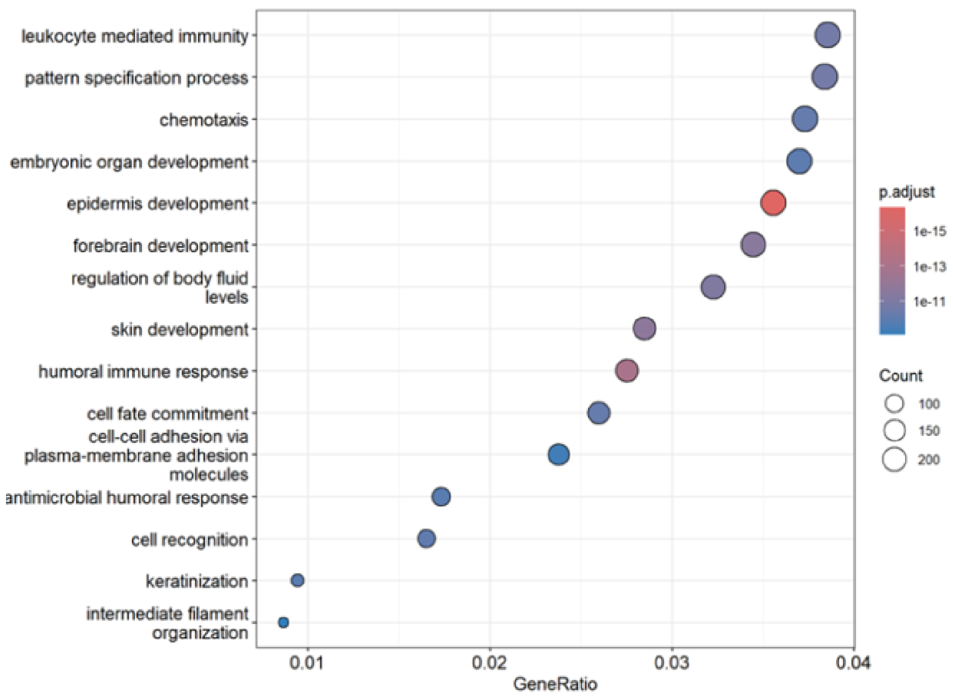

Figure 4. Gene Ontology enrichment analysis.

Top enriched biological processes among differentially expressed genes in TCGA-LUSC tumors.

Construction of the prognostic gene expression signature

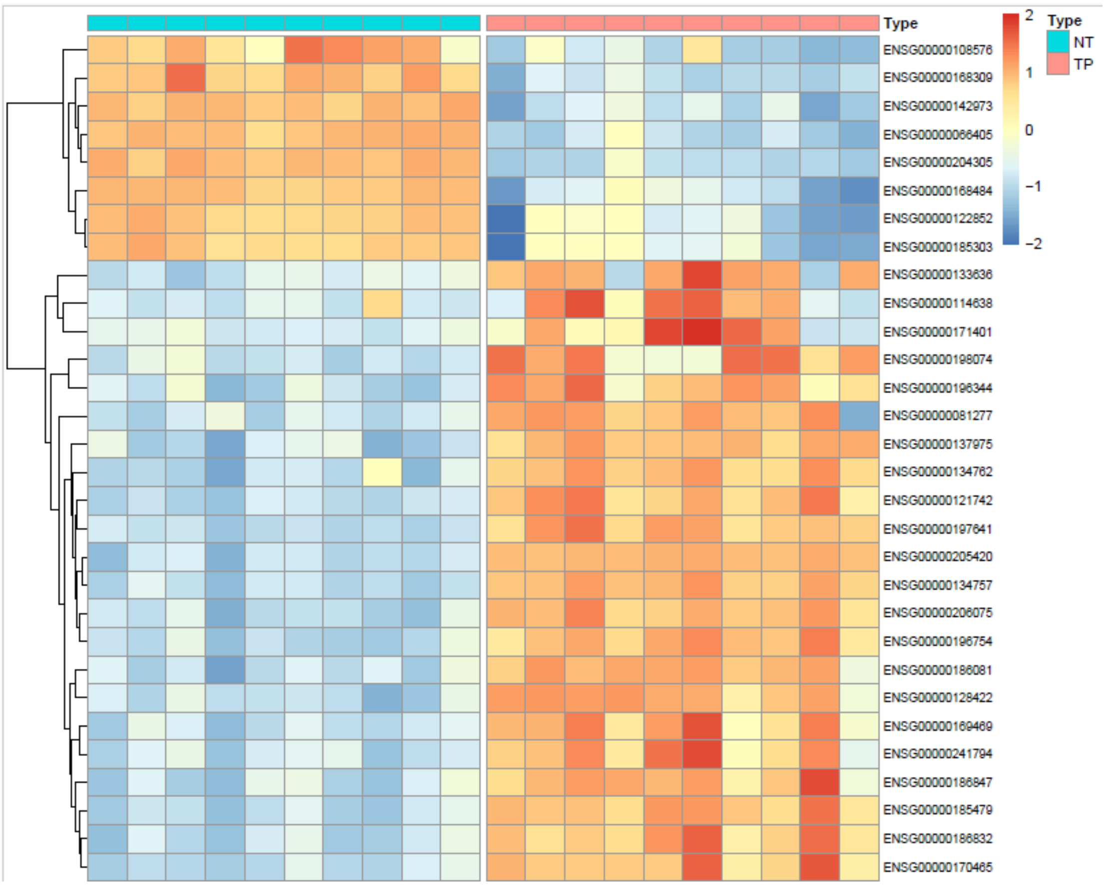

Figure 5. Heatmap of differentially expressed genes distinguishing tumor and normal samples.

Z-score normalized expression of selected signature genes across tumor (TP) and normal (NT) samples.

Candidate biomarker genes were selected from the set of significantly differentially expressed genes. Genes were ranked according to statistical significance and effect size (log2 fold change).

For each patient, a signature score was calculated as the mean z-score of normalized expression values across the selected signature genes.

Patients were stratified into HIGH and LOW signature groups using the median signature score within the cohort.

Survival analysis

The prognostic relevance of the gene expression signature was evaluated using survival analysis.

Kaplan–Meier survival curves were constructed to compare overall survival between signature-defined patient groups.

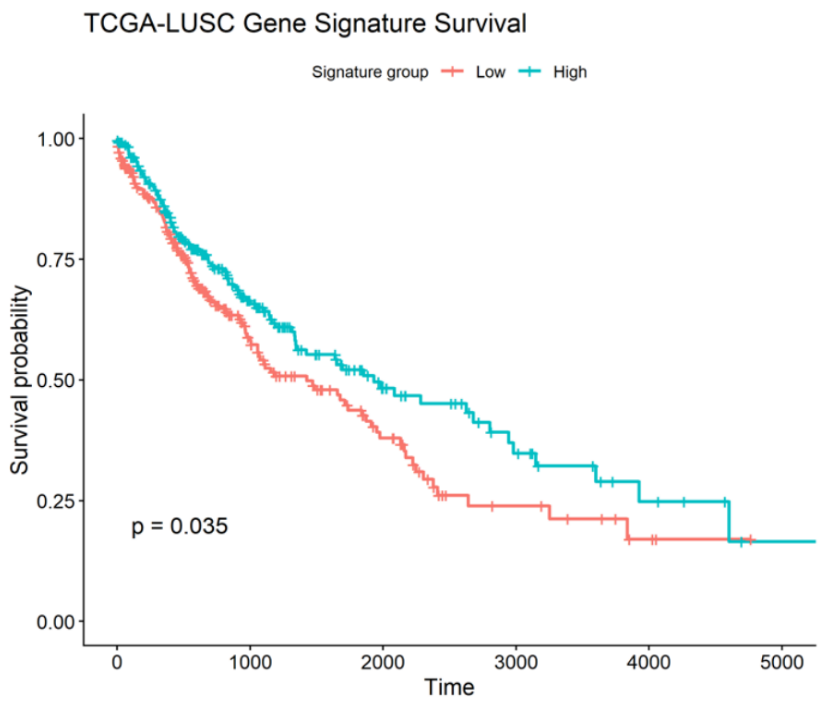

Figure 6. Prognostic gene expression signature.

Kaplan–Meier survival curves showing stratification of TCGA-LUSC patients based on the transcriptomic signature.

Statistical significance between groups was assessed using the log-rank test.

To evaluate the independent prognostic value of the signature, a Cox proportional hazards regression model was fitted including:

- signature score
- patient age
- AJCC pathological stage

Model performance was evaluated using the concordance index (C-index) and likelihood ratio tests.

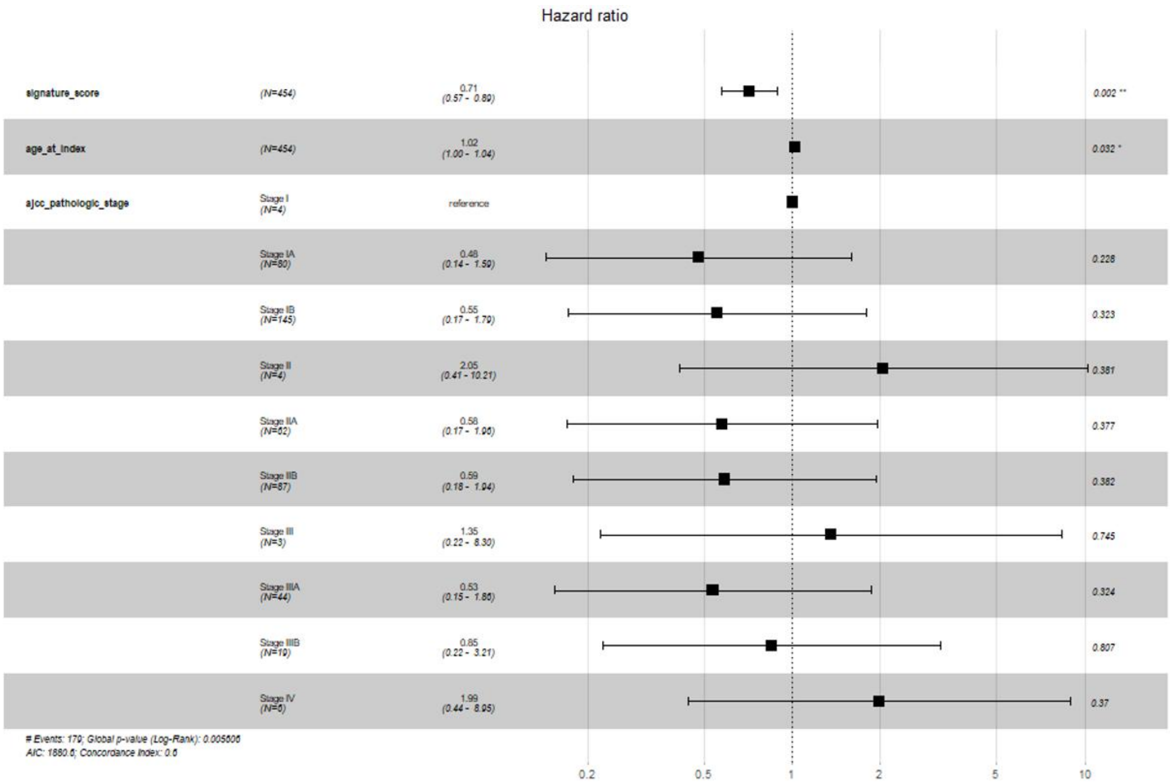

Figure 7. Multivariate Cox proportional hazards analysis.

Forest plot showing hazard ratios for the transcriptomic signature and clinical covariates.

External validation

To assess the generalizability of the TCGA-derived gene signature, external validation was performed using an independent cohort from the Gene Expression Omnibus (GEO) database (dataset GSE33479).

Expression profiles from the validation dataset were processed and mapped to the TCGA-derived signature genes. Signature scores were calculated using the same procedure as in the TCGA cohort.

Principal component analysis and expression score comparisons were used to evaluate the ability of the signature to distinguish tumor and normal samples in the independent dataset.

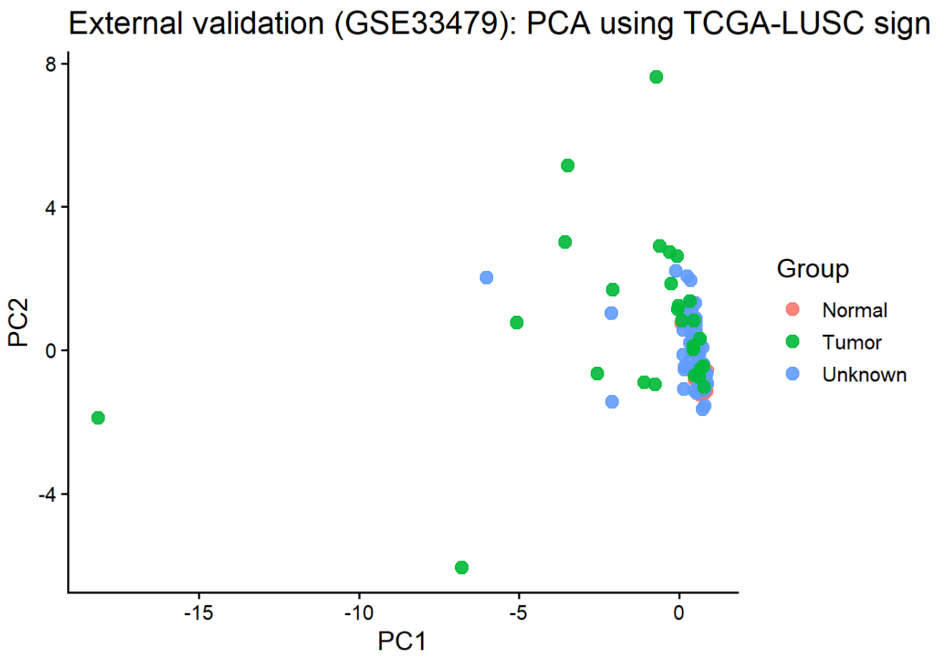

Figure 8. External validation in the GEO dataset (GSE33479).

PCA using TCGA-derived signature genes in the independent GEO cohort.

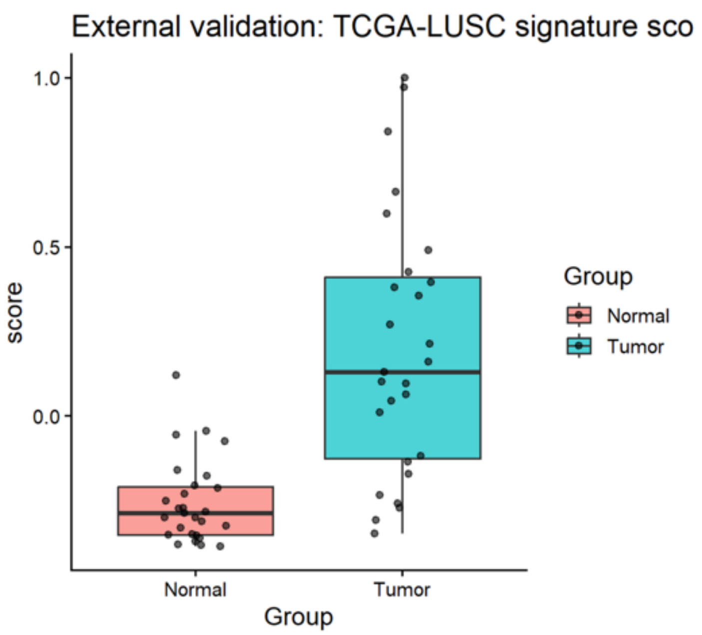

Figure 9. Signature score comparison in GEO dataset.

Distribution of the transcriptomic signature score between tumor and normal samples.

Software and reproducibility

All analyses were performed in R (version ≥ 4.2).

Key R packages used in the analysis included:

- TCGAbiolinks
- DESeq2
- clusterProfiler
- survival
- survminer
- ggplot2
- pheatmap

The complete analysis pipeline is fully reproducible and available as open-source code at:

https://github.com/ag48665/tcga-lusc-biomarker-analysis

The workflow can be executed using a single command (run_analysis.R) that downloads data, performs preprocessing, and generates all results and figures.

Discussion

Our analysis demonstrates that transcriptomic profiling of TCGA-LUSC tumors reveals clear and reproducible differences between tumor and normal lung tissue. Principal component analysis showed strong separation of tumor and normal samples, indicating large-scale transcriptional reprogramming in lung squamous cell carcinoma.

Functional enrichment analysis highlighted biological processes consistent with the known pathology of LUSC, including keratinization, epithelial differentiation, and epidermal development. These pathways reflect the squamous epithelial lineage of these tumors and confirm that transcriptomic analysis captures key aspects of tumor cell identity.

In addition to tumor differentiation programs, we observed strong enrichment of immune-related pathways, including leukocyte-mediated immunity and chemotaxis. This finding is consistent with the known importance of tumor–immune microenvironment interactions in lung cancer biology.

Importantly, we identified a multi-gene transcriptional signature capable of stratifying patients by overall survival. The prognostic effect remained significant after adjustment for clinical covariates, suggesting that transcriptomic signals capture biological risk factors not fully explained by tumor stage alone.

Together, these results demonstrate that transcriptome-wide analysis of RNA-seq data can identify biologically meaningful gene expression programs and clinically relevant prognostic signatures in lung squamous cell carcinoma.

Limitations

This study has several limitations. First, the analysis was performed using retrospective data obtained from public cancer genomics repositories such as TCGA. Although TCGA provides a large and well-characterized dataset, retrospective studies may be affected by cohort-specific biases and clinical heterogeneity.

Second, the study relies on bulk RNA-sequencing data, which represent average gene expression across all cells within a tumor sample and therefore do not capture cell-type–specific transcriptional heterogeneity within the tumor microenvironment.

Third, although the prognostic gene signature was evaluated using survival analysis and externally assessed in an independent dataset, further validation in additional cohorts and prospective clinical studies would be required before clinical application.

Finally, the gene signature was derived from transcriptomic differences between tumor and normal samples and may be influenced by dataset-specific characteristics. Additional validation and integration with other molecular data types could further improve the robustness and clinical applicability of the model.

Data Availability

The datasets analyzed in this study are publicly available from established cancer genomics repositories. RNA-sequencing data for lung squamous cell carcinoma (TCGA-LUSC) were obtained from The Cancer Genome Atlas (TCGA) through the Genomic Data Commons (GDC) data portal.

TCGA data can be accessed at:
https://portal.gdc.cancer.gov/

External validation data used in this study were obtained from the Gene Expression Omnibus (GEO) repository under accession number GSE33479.

GEO data are available at:
https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE33479

All datasets used in this study are publicly available, de-identified, and do not require additional ethical approval.

Code Availability

All code used for data acquisition, preprocessing, differential expression analysis, pathway enrichment analysis, survival modeling, and visualization is publicly available in the project repository:

https://github.com/ag48665/tcga-lusc-biomarker-analysis

The repository contains a fully reproducible bioinformatics workflow implemented in R, including scripts for:

- TCGA data download using TCGAbiolinks
- RNA-seq differential expression analysis using DESeq2
- functional enrichment analysis using clusterProfiler
- survival analysis using survival and survminer
- data visualization using ggplot2 and pheatmap

The complete pipeline can be executed using the main script:

run_analysis.R which automatically downloads the required data and reproduces all results and figures presented in this study.

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